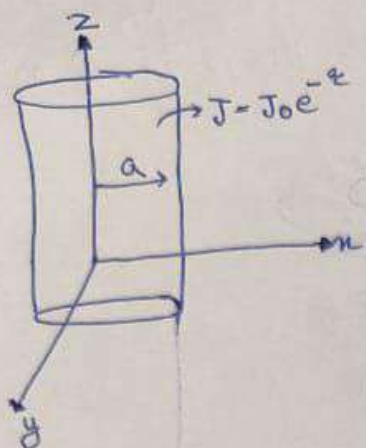


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Assignment - 2

Ans 1



① Magnetic field H for
 $x \in [0, a]$

$$\oint H \cdot d\mathbf{l} = I_{enc} = \int_0^x J dx$$

$$= \int_0^x \int_0^{2\pi} \int_0^a J_0 e^{-x'} r' d\phi dx'$$

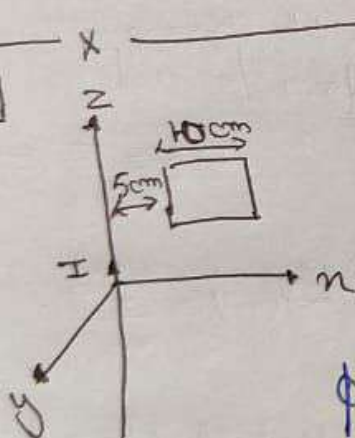
$$H \cdot 2\pi x = 2\pi J_0 \int_0^x x' e^{-x'} dx'$$

$$H = J_0/x (1 - x e^{-x} - e^{-x})$$

② Now $x \gg a \Rightarrow H = J_0/a (1 - a e^{-a} - e^{-a})$

③ Now $x > a$, we get $\Rightarrow \boxed{\frac{J_0}{x} (1 - a e^{-a} - e^{-a}) = H}$

Ans 2



① $\mathcal{E}MF = -\frac{\partial \Phi}{\partial t}$ $\Phi = \int B \cdot d\mathbf{A}$

$$B = \frac{\mu_0 I(t)}{2\pi r} \rightarrow \Phi = \int_x^{x+L} \frac{\mu_0 I(t)}{2\pi r} L dr$$

$$\Phi = \frac{\mu_0 I(t) L}{2\pi} \ln\left(\frac{x+L}{x}\right) \quad [\text{put } L=10 \text{ and } x=5]$$

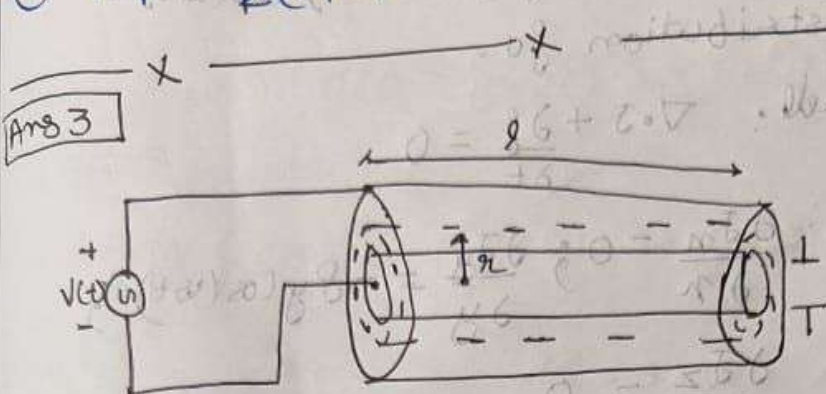
$$\Rightarrow \Phi = \frac{\mu_0 I(t) \times 5}{2\pi} \ln(3)$$

$$\text{Now } \mathcal{E}MF = -\frac{\partial \Phi}{\partial t} = -\frac{\mu_0 5 \ln(3)}{2\pi} \frac{\partial (\cos(2\pi \times 10^4 t))}{\partial t}$$

$$= \mu_0 \times 5 \times 10^4 \ln(3) \sin(2\pi \times 10^4 t)$$

$$\boxed{\mathcal{E}MF = 0.00698 \sin(2\pi \times 10^4 t) \text{ Volt}}$$

$$\textcircled{b} I_{\text{loop}} = \frac{V}{R} = 0.0069 \sin(2\pi 10^4 t) \text{ A}$$



$$C = \frac{2\pi\epsilon_0 l}{\ln(b/a)}$$

①

Substituting the values:

$$C = \frac{2 \times (8.854 \times 10^{-12} \times 9) \times 0.06}{\ln(0.04/0.005)}$$

$$= \frac{2\pi \times 7.9686 \times 10^{-11} \times 0.06}{\ln(2)}$$

$$C \approx 4.33 \times 10^{-11} \text{ Farad}$$

② Displacement Current I_d :-

$$Q(t) = C \times V(t)$$

$$\& V(t) = 50 \sin(120\pi t);$$

$$Q(t) = 4.33 \times 10^{-11} \times 50 \times \sin(120\pi t)$$

$$I_d = \frac{\partial Q(t)}{\partial t} = 4.33 \times 5 \times 10^{-10} \times 120\pi \times \cos(120\pi t)$$

$$= 2.765 \times 10^{-7} \cos(120\pi t)$$

Ans 4

$$\vec{J}(x, y, z, t) = (2\hat{x} - 4y^2\hat{y} + 2xz\hat{z}) \cos(\omega t)$$

• charge distribution ρ_0 .

Eqn of Maxwell: $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$

$$\vec{J} = \vec{J} \quad \therefore \frac{\partial J_x}{\partial x} = 0; \frac{\partial J_y}{\partial y} = -8y \cos(\omega t);$$

$$\frac{\partial J_z}{\partial z} = 0$$

$$\therefore \frac{\partial J_y}{\partial y} \nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$\Rightarrow -8y \cos \omega t + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \boxed{\rho = \frac{8y}{\omega} \sin(\omega t)}$$

Ans 5

We need to find ω for $J_c = J_d$ in lossy medium i.e. $\sigma \neq 0$

$$\Rightarrow J_c = \sigma E$$

$$J_d = \epsilon \frac{\partial E}{\partial t}$$

[where E varies sinusoidally with time as $E(t) = E_0 e^{j\omega t}$]

$$\frac{\partial E}{\partial t} = j\omega E$$

$$J_d = \epsilon j\omega E$$

Now $|J_c| = |J_d|$

$$\omega \epsilon E = \sigma E$$

$$\Rightarrow \omega = \sigma / \epsilon = 2\pi f \Rightarrow \boxed{f = \sigma / 2\pi \epsilon}$$

- Ans 6
- Dry Soil $\epsilon_r = 2.5$ $\sigma = 10^{-4} \text{ S/m}$
 - Fresh water $\epsilon_r = 80$ $\sigma = 10^{-3} \text{ S/m}$

So, for a given ω , the smaller σ/ϵ is a better insulator as fresh water is a better insulator.

Ans 7 Given $E = 3 \sin(10^7 \pi t - 0.2 \pi x) \hat{y} + 4 \cos(10^7 \pi t - 0.2 \pi x) \hat{z} \text{ V/m}$

(a) Wavelength: Comparing the above eqn with the general form of a propagating wave, we get

$$\beta = 0.2 \pi \quad \lambda = \frac{2 \pi}{\beta} \Rightarrow \boxed{\lambda = 10 \text{ m}}$$

(b) From the eqn $\omega = 10^7 \pi$
 Now $v = \lambda f = \lambda / \beta \Rightarrow v = \frac{\pi \times 10^7}{0.2 \pi} = 5 \times 10^7 \text{ m/s}$

Speed in medium is dependant on medium

properties i.e. $v = \frac{c}{\sqrt{\epsilon_r}} \Rightarrow \epsilon_r = \left(\frac{c}{v}\right)^2 = \left(\frac{3 \times 10^8}{5 \times 10^7}\right)^2 = 36 = \epsilon_r$

(c) Intrinsic impedance of medium:

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} = \frac{\mu_0}{\sqrt{\epsilon_r} \epsilon_0} = \frac{4 \pi \times 10^{-7}}{6 \times 10^{-9}} = 62.83 \Omega$$

(d) Magnetic field $\vec{H} = \frac{1}{\eta} (\hat{k} \times \vec{E})$ [$\hat{k}$ is direction of wave propagation]

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$$H = \frac{1}{62.83} \hat{n} \times [3 \sin(10^7 \pi t - 0.2 \pi n) \hat{y} + 4 \cos(10^7 \pi t - 0.2 \pi n) \hat{z}]$$

$$\Rightarrow H = \frac{1}{62.83} [4 \cos(10^7 \pi t - 0.2 \pi n) \hat{y} - 3 \sin(10^7 \pi t - 0.2 \pi n) \hat{z}]$$

Ans 8. $H = 6 \cos(2z) \sin(2 \times 10^7 t - 0.1 \pi n) \hat{y} \text{ A/m}$
 $\mu_0 = 1$

(a) Electric Field: $\nabla \times E = \frac{\partial B}{\partial t}$

$$B = \mu_0 H$$

$$\Rightarrow B = 4 \pi \times 10^{-7} \times 6 \cos(2z) \sin(2 \times 10^7 t - 0.1 \pi n)$$

$$\Rightarrow \frac{\partial B}{\partial t} = 4 \pi \times 6 \times 2 \cos(2z) \cos(2 \times 10^7 t - 0.1 \pi n) \hat{y}$$

$$\nabla \times E = \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) \hat{x} + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial n} \right) \hat{y} + \left(\frac{\partial E_y}{\partial n} - \frac{\partial E_n}{\partial y} \right) \hat{z}$$

$\Rightarrow$ Equating the y-components:

$$\frac{\partial E_n}{\partial z} - \frac{\partial E_z}{\partial n} = 48 \pi \cos(2z) \cos(2 \times 10^7 t - 0.1 \pi n)$$

Since, wave is propagating along n-axis, the E will have only $\hat{z}$ component so $E_n = 0$

$$\Rightarrow -E_z = \int 48 \pi \cos(2z) \cos(2 \times 10^7 t - 0.1 \pi n) \partial n$$

$$E_z = 480 \pi \cos(2z) \sin(2 \times 10^7 t - 0.1 \pi n) \hat{z}$$

⑥ Displacement current density:

$$\vec{J}_d = \frac{\partial \vec{E}}{\partial t} \quad \epsilon = \epsilon_0 \epsilon_r$$

$$\frac{1}{\sqrt{\mu \epsilon}} = \frac{2 \times 10^7}{0.1} \quad , \quad \frac{1}{\sqrt{\mu_0 \epsilon_r \epsilon_0}} = \frac{3 \times 10^8}{\sqrt{\epsilon_r}} = 2 \times 10^8 \Rightarrow \boxed{\epsilon_r = 9/4}$$

$$\text{So, } J_d = \frac{9}{4} \epsilon_0 480 \pi \cos(2z) \frac{\partial (\sin(2\pi \times 10^7 t - 0.1\pi))}{\partial t}$$

$$\Rightarrow \boxed{\vec{J}_d = 0.6 \cos(2z) \cos(2 \times 10^4 t - 0.1\pi) \hat{z}}$$

⑦ Charge Density $\rho \rightarrow \nabla \cdot \epsilon = \rho / \epsilon_0$

$$\text{Since } t = t_z \quad \nabla \cdot \epsilon = \frac{\partial \epsilon_z}{\partial z} = -480 \pi \times 2 \sin(2z) \sin(2 \times 10^7 t - 0.1\pi)$$

$$\text{So } \rho = \nabla \cdot \epsilon \epsilon_0 \epsilon_r = -1.28 \sin(2z) \sin(2 \times 10^7 t - 0.1\pi)$$